

# Mohammad VazirPanah



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Nationality: Iranian · DOB: 23 Apr 1994

## Research Interests

Embedded Systems, Robotics, HPC, RISC-V Architectures, Real-Time Heterogeneous Platforms

## Education

### M.Sc. in Computer Engineering — Computer Systems Architecture

Sep 2019 – Oct 2022

Shahid Beheshti University, Tehran, Iran

- GPA: 3.79 / 4.00
- Thesis: *A Framework for Modeling Self-Aware Cyber-Physical Systems with SystemC*
- Supervisor: Dr. Seyed-Hosein Attarzadeh-Niaki

### B.Sc. in Computer Engineering — Hardware Engineering

Sep 2012 – May 2016

Birjand University, Birjand, Iran

- GPA: 3.40 / 4.00
- Project: *Simulation and Programming the Alpha Rex Human-Robot Produced by Lego Company*

## Work Experience

### Research Fellow

2023 – 2026

University of Salerno — Dept. of Computer Science, ISISLab, Salerno, Italy

- Scientific Responsible: Prof. Biagio Cosenza
- Focus on High-Performance Computing and RISC-V architectures
- Contributed to [LibreRT](#): developing and deploying real-time applications on embedded CPUs and GPUs
- Evaluated compiler performance (GCC, LLVM) on RISC-V vector boards
- Deployed SoC boards; managed experiments with Git, Docker, CI; documented setups for reproducibility

### Instructor

2013 – 2015

Sepehr Educational Institutions, Kashmar, Iran

- Delivered theoretical and practical instruction in C++ programming
- Taught ICDL (International Computer Driving License) skills
- Taught CorelDRAW and Adobe FreeHand, focusing on vector graphics

### Education Intern

2014 – 2015

Abooreyhan Company, Kashmar, Iran

- Hardware/software troubleshooting, diagnostics, and component replacement
- Assembled and configured PCs; software installation and system setup

## Academic Experience

### Teaching Assistantships

|                                                            |               |            |
|------------------------------------------------------------|---------------|------------|
| <i>Design &amp; Analysis of Real-Time Embedded Systems</i> | SBU           | 2021       |
| <i>System-on-Chip</i>                                      | SBU           | 2020, 2021 |
| <i>Advanced Computer Architecture</i>                      | SBU           | 2021       |
| <i>Fundamentals of Embedded &amp; Real-Time Systems</i>    | SBU           | 2020       |
| <i>Computer Architecture</i>                               | Birjand Univ. | 2016       |

### Laboratory Memberships

- ISISLab — HPC Group, University of Salerno, Italy
- SBU Cyber-Physical Systems Lab Research Team, Tehran, Iran

## Publications

- 2026** **LibreRT: Portable Heterogeneous Real-Time Programming for the Embedded Computing Continuum**  
*B. Cosenza, F. Terraneo, G. Agosta, L. Carpentieri, D. Cattaneo, M. VazirPanah, N. Nicolosi*  
23rd ACM International Conference on Computing Frontiers (CF), Special Sessions · 2026
- 2025** **A Performance Analysis of Autovectorization on RVV RISC-V Boards**  
*L. Carpentieri, M. VazirPanah, B. Cosenza*  
PDP — 33rd Euromicro International Conference on Parallel, Distributed, and Network-Based Processing, IEEE · 2025
- 2023** **System-Level Modeling of Dynamic Applications with Scenario-Aware Dataflow Graphs**  
*S.-H. Attarzadeh-Niaki, M. VazirPanah*  
Journal of Innovations in Computer Science and Engineering (JICSE), Shahid Beheshti University · 2023
- 2022** **ROS-Based Co-Simulation for Formal Cyber-Physical Robotic System Design**  
*M. VazirPanah, S.-H. Attarzadeh-Niaki, A. Salimi-Badr*  
27th International Computer Conference, Computer Society of Iran (CSICC), IEEE · 2022

## Selected Projects

- LibreRT — Portable Heterogeneous Real-Time Programming for the Embedded Computing Continuum** 2023 – 2026  
*Researcher, Italian PRIN 2022-funded project*
  - Developed and evaluated real-time applications for heterogeneous embedded CPU/GPU platforms
  - Implemented solutions using SYCL, OpenMP, OpenCL, and RISC-V boards
- RISC-V Benchmark Suite** 2025  
*Open-source project [GitHub](#)*
  - Designed and implemented benchmarks for RISC-V and RVV architectures
  - Evaluated performance across multiple GCC and LLVM compiler versions
- Self-Aware CPS Modeling Framework** 2022  
*M.Sc. Thesis [Open-source project GitHub](#) based on [ForSyDe-SystemC Modeling Framework](#)*
  - Designed and implemented a self-aware cyber-physical system model using ForSyDe-SystemC and ROS-Gazebo co-simulation
  - Developed an SADF-based model and contributed extensions to the ForSyDe-SystemC library
- Railroad System Verification with NuSMV** 2020  
*Course Project, M.Sc. Design and Analysis of Real-Time Embedded Systems*
  - Modeled and verified a railroad control system using NuSMV
  - Checked safety and liveness properties through formal verification
- FIR Filter and Tiny YOLO v1 on FPGA** 2019  
*Course Project, M.Sc. System-on-Chip, DE0-Nano / DE10 Intel FPGA Board*
  - Designed and implemented a FIR filter IP core in Quartus with an Avalon-MM interface
  - Implemented real-time Tiny YOLO v1 object detection on FPGA under resource constraints

## Technical Skills

|                             |                                                                     |
|-----------------------------|---------------------------------------------------------------------|
| <b>Programming</b>          | C, C++, Python, MATLAB, Assembly                                    |
| <b>Parallel Models</b>      | OpenMP, OpenCL, SYCL                                                |
| <b>Hardware / Embedded</b>  | FPGA, RISC-V, VHDL, Verilog, SystemC, FreeRTOS                      |
| <b>Simulation &amp; EDA</b> | Gem5, QEMU, OVPsim, Quartus, ModelSim, MATLAB-Simulink, Ansys SCADE |
| <b>Robotics</b>             | ROS, Gazebo                                                         |
| <b>DevOps &amp; Tools</b>   | Git, Docker, Linux, $\LaTeX$ , OpenModelica, NuSMV, UPPAAL          |

## Honours and Awards

### National R&D Research Grant Winner

2020

*R&D Center, Mobile Telecommunication Company of Iran (MCI)*

- Awarded among competing Master's and PhD students in a national grant competition

### Top Computer Engineering Student

2016

*Birjand University*

- Ranked in the top 15% of B.Sc. Computer Engineering graduates

## Conference Participation

### 33rd Euromicro International Conference on Parallel, Distributed, and Network-Based Processing (PDP)

Mar 2025

*Co-author and attendee — Turin, Italy*

- Co-author of the paper: *A Performance Analysis of Autovectorization on RVV RISC-V Boards*

### 27th International Computer Conference, Computer Society of Iran (CSICC)

Feb 2022

*Presenter — Sharif University of Technology, Tehran, Iran*

- Presented the paper: *ROS-Based Co-Simulation for Formal Cyber-Physical Robotic System Design*

## Language Skills

|         |                                                      |
|---------|------------------------------------------------------|
| Persian | Native                                               |
| English | Proficient — (Listening, Reading, Writing, Speaking) |
| Italian | Basic — (Listening, Reading, Writing, Speaking)      |

## References

### Prof. Biagio Cosenza

*Research Supervisor*

University of Salerno, Italy

[bcosenza@unisa.it](mailto:bcosenza@unisa.it)

### Prof. Vittorio Scarano

*Head of ISISLab*

University of Salerno, Italy

[vitsca@unisa.it](mailto:vitsca@unisa.it)

### Dr. S.-H. Attarzadeh-Niaki

*M.Sc. Supervisor*

Shahid Beheshti University, Iran

[h\\_attarzadeh@sbu.ac.ir](mailto:h_attarzadeh@sbu.ac.ir)